

JOINT KINEMATICS DURING STEEP UPHILL WALKING

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Introduction: Walking on steep inclines necessitates adaptations in joint kinematics to maintain stability and efficiency. As extreme elevation ultra-trail races [1], such as the Hardrock 100 ($\pm 20,000\text{m}$) and the Ultra-Trail du Mont Blanc ($\pm 24,000\text{m}$) gain in popularity, athletes question if the same performance benefits of running in stiffer shoes on level ground [2] will extend to steep slopes. Previous studies have highlighted the importance of understanding joint kinematics in response to graded locomotion [3,4], but are limited in their slope ranges ($< \pm 10^\circ$), joint-specific analysis, or considerations of the metatarsal-phalangeal joints (MTP). Papachatzis et al. quantified mechanical work and joint kinematics during walking on inclines up to $\pm 10^\circ$ but did not extend their analysis beyond this range [3]. Whiting et al. investigated gait parameters during uphill walking at a 30° incline, identifying changes in gait characteristics without joint-specific analysis [5]. In contrast, Swanson and Caldwell examined joint kinematics at inclines up to 16.7° , focusing on ankle, knee, and hip kinematics but did not consider the MTP joint [4]. Therefore, there is still a gap in our understanding of leg joint kinematics on steep slopes. Our study aimed to address the effects of increasing incline on MTP and ankle joint kinematics. By examining joint kinematics across a wide range of inclines, we provide insights that could inform terrain-specific footwear design and enhance trail running or hiking performance. We hypothesized that as slope increased, there would be a corresponding increase in MTP and ankle joint dorsiflexion angles.

Methods: 12 healthy participants (21.8 ± 2.9 years, 171.5 ± 7.8 cm, 63.9 ± 7.2 kg) completed 1-minute walking trials at inclines of 0° , 10° , 20° , or 30° and a constant speed of 0.6 m/s. To approximate walking barefoot while limiting the risk of injury due to friction, participants wore minimalist shoes (Vibram FiveFingers KSO EVO, Vibram Corporation, North Brookfield MA, US). We placed twenty-two reflective markers on relevant anatomical landmarks at the pelvis, the left thigh, shank, and foot. A custom-built treadmill capable of inclines ranging from 5° to 45° , prepared with a layer of high-friction grip tape was used for inclined trials [6]. A second treadmill was used for level (0°) conditions (Treadmetrix, Park City UT, US). During the trials, motion capture data (Qualisys Inc., Gothenburg, Sweden) were continuously collected at 200 Hz. A generic skeletal model was scaled to each participant in OpenSim 4.4 before the Inverse Kinematics tool was used to calculate joint angles [7]. Walking trials were segmented into strides based on the anterior-posterior location of the first head MTP joint marker. Ankle and MTP joint angles were averaged across strides for each participant and were used to calculate groupwise averages and standard deviations across conditions. Peak joint dorsiflexion angle was determined as the absolute maximum for a given condition. Midstance MTP angle was calculated as the average from 20 to 40% of the gait cycle. We used one-way ANOVAs to evaluate peak MTP, midstance MTP, and peak ankle angle across inclines. Only 11 of the 12 collected participants were processed due to challenges with the scaling process.

Results & Discussion: Increasing incline had a significant effect on peak MTP dorsiflexion angle ($p = .005$), midstance MTP angle ($p = .004$), and peak ankle dorsiflexion angle ($p < 0.001$) (Figure 1). All angles increased with increasing incline, supporting our hypothesis. There was considerable participant variation for both joints, particularly at 30° at which midstance MTP dorsiflexion angle was negatively correlated to peak ankle dorsiflexion angle ($r^2 = 0.79$), highlighting the different joint angle strategies participants utilized to walk up the steep incline successfully.

Significance: Our results show that increasing incline results in significant increases in MTP and ankle joint dorsiflexion angles. Further investigation into if and how these findings vary when athletes choose shoes with different properties will help inform shoe design and deepen our understanding of locomotion on steep inclines.

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References: [1] Vernillo et al. (2016), *Sports Med* 47(4); [2] Hoogkamer et al. (2017), *Sports Med* 48(4); [3] Papachatzis et al. (2023), *PLOS ONE* 18(9); [4] Swanson et al. (2000), *Med & Sci in Sports & Exercise* 32(6); [5] Whiting et al. (2020), *Euro Journal of Appl Phys* 120(10); [6] Giovanelli et al. (2016), *Journal of Appl Phys* 120(3). [7] A. Seth et al. (2018), *PLOS Comput. Biol.* 14(7)

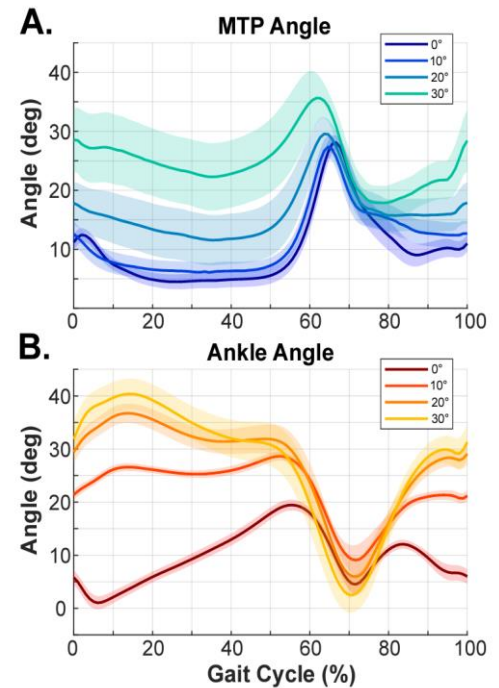


Figure 1: Group average (solid) and standard deviation (shaded) for MTP angle (A) and ankle angle (B) across incline.